

Shayan Majidy, Ph.D.

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RESEARCH POSITIONS

2024– [Banting Postdoctoral Fellow](#), Department of Physics, Harvard University
Faculty Advisors: Misha Lukin, Joshua and Beth Friedman University Professor
Michael Gullans, NIST Physicist & QuICS Fellow

EDUCATION

2019–2024 **Ph.D. in Physics**, University of Waterloo
Faculty Advisors: Raymond Laflamme, late Professor of Physics
Nicole Yunger Halpern, NIST Physicist & QuICS Fellow
Thesis: Effects of Noncommuting Charges in Quantum Information and Thermodynamics
Funding: [Vanier Scholarship](#), 2021-2024

2018–2019 **M.Sc. in Physics**, University of Waterloo
Faculty Advisors: Raymond Laflamme, late Professor of Physics

2011–2015 **B.Sc. in Theoretical Physics**, University of Guelph

PUBLICATIONS

Textbooks

- **S. Majidy**, C. Wilson, and R. Laflamme, “[Building quantum computers: A practical introduction](#),” Cambridge University Press, (2024).

Journal Articles

1. D. Bluvstein, A. Geim, . . . , **S. Majidy**, . . . , and M.D. Lukin, “[A fault-tolerant neutral-atom architecture for universal quantum computation](#),” *Nature* 649, 39–46 (2026).
2. **S. Majidy**, D. Hangleiter, and M. Gullans, “[Scalable and fault-tolerant preparation of encoded k-uniform states](#),” *Phys. Rev. A* 112, 042409 (2025).
3. **S. Majidy** “[Noncommuting charges can remove non-stationary quantum many-body dynamics](#),” *Nat. Comm.* 15, 8246 (2024).
4. **S. Majidy**, W. F. Braasch, Jr., A. Lasek, T. Upadhyaya, A. Kalev, and N. Yunger Halpern, “[Noncommuting conserved charges in quantum thermodynamics and beyond](#),” *Nat. Rev. Phys.* 5, 689–698 (2023).
5. **S. Majidy**, U. Agrawal, S. Gopalakrishnan, A. Potter, R. Vasseur, and N. Yunger Halpern “[Critical phase and spin sharpening in SU\(2\)-symmetric monitored quantum circuits](#),” *Phys. Rev. B* 108, 054307 (2023).
6. **S. Majidy** “[A unification of the coding theory and OAQEC perspective on hybrid codes](#),” *Int. J. Theor. Phys.* 62.8: 177 (2023).
7. **S. Majidy**, A. Lasek, D. A. Huse, and N. Yunger Halpern, “[Non-abelian symmetry can increase entanglement entropy](#),” *Phys. Rev. B*, 107, 045102 (2023).
8. N. Yunger Halpern and **S. Majidy**, “[How to build hamiltonians that transport noncommuting charges in quantum thermodynamics](#),” *npj Quantum Information* 8, 10 (2022)
9. **S. Majidy**, J. J. Halliwell, and R. Laflamme, “[Detecting violations of macrorealism when the original Leggett-Garg inequalities are satisfied](#),” *Phys. Rev. A* 103, 062212 (2021)
10. **S. Majidy**, H. Katiyar, G. Anikeeva, J. Halliwell, and R. Laflamme, “[Exploration of an augmented set of Leggett-Garg inequalities using a noninvasive continuous-in-time velocity measurement](#),” *Phys. Rev. A*, 100, 042325 (2019).

Preprints

12. J. M. Koh, A. Gong, A. C. Diaconu, D. B. Tan, A. A. Geim, M. J. Gullans, N. Y. Yao, M. D. Lukin, and **S. Majidy** “[Entangling logical qubits without physical operations](#),” arXiv:2601.20927 (2026).

13. M. Bilokur, S. Gopalakrishnan, and **S. Majidy** “[Thermodynamic limitations on fault-tolerant quantum computing](#),” arXiv:2411.12805 (2024).
14. **S. Majidy** “[Addressing misconceptions in university physics: A review and experiences from quantum physics educators](#),” arXiv:2405.20923 (2025).

TALKS

Conference Talks & Colloquia (11)

Entangling logical qubits without physical operations

- Princeton Quantum Technology Conference 2026, Princeton, New Jersey (Oct 22, 2026) **[Invited]**
- [TQC 2025](#), Sherbrooke, Québec (Aug 13, 2025)

Fault-Tolerant Quantum Computing: From Fundamental Limits to Scalable Architectures

- University of Alberta Colloquium, University of Alberta, Edmonton, Alberta (Apr 16, 2026)
- UBC-Okanagan Colloquium, UBC-Okanagan, Kelowna, BC (Mar 20, 2026)
- Institut Quantique Colloquium, Université de Sherbrooke, Sherbrooke, Québec (Jan 15, 2026)

Architectural mechanisms of a universal fault-tolerant quantum computer

- [Year of Quantum Across Canada](#), Waterloo, Ontario (Oct 7, 2025) [\[Recording\]](#)

Non-Abelian symmetry can increase entanglement entropy

- [CQIQC-X](#), University of Toronto, Ontario (Aug 26, 2024). [\[Recording\]](#)
- [Raymond Laflamme’s 60th Birthday Conference](#), University of Waterloo, Ontario (Jul 19, 2023).
- [RQS annual workshop](#), University of Maryland, Maryland (June 22, 2023).
- [Quantum Non-Markovianity 2022](#), Online, (Dec 8, 2022)

Noncommuting charges: Bridging theory to experiment

- [Information Engines at the Frontiers of Nanoscale Thermodynamics 2022](#), Telluride Science Research Center, Colorado (July 22, 2022).

Invited Seminars (21)

Entangling logical qubits without physical operations

- [QASAR Seminar](#), Inria, Online (Apr 13, 2025)
- [QuICS Seminar](#), University of Maryland, College Park, Maryland (Mar 4, 2026) [\[Recording\]](#)
- PI Quantum Information Seminar, Perimeter Institute, Waterloo, Ontario (Jan 7, 2026) [\[Recording\]](#)

Architectural mechanisms of a universal fault-tolerant quantum computer

- [CQIQC Seminar](#), University of Toronto, Toronto, Ontario (Oct 3, 2025) [\[Recording\]](#)

A noncommuting charge puzzle

- [PI Quantum Information Seminar](#), Perimeter Institute, Waterloo, Ontario (Jan 8, 2025) [\[Recording\]](#)
- Harvard Quantum Initiative Seminar, Cambridge, Massachusetts (Jun 28, 2024)

Noncommuting charges can increase entanglement and induce critical dynamics

- [Yale Quantum Institute Seminar](#), Yale, Connecticut (Jan 16, 2024)
- PI Quantum Information Seminar, Perimeter Institute, Ontario (Nov 29, 2023) [\[Recording\]](#)
- Princeton Centre for Theoretical Physics seminar organized by Biao Lian, Princeton, New Jersey (Sept 22, 2023).

Monitored Quantum Circuits with Noncommuting Conserved Quantities

- [Qiskit Seminar](#), IBM, Online (Sept 15, 2023). [\[Recording\]](#)

Non-Abelian symmetry can increase entanglement entropy

- [InfoQ Seminar](#), Institut Quantique, Quebec (Mar 28, 2023).
- [Special INTRIQ/CPM Seminar](#), McGill University, Quebec (Mar 24, 2023).
- Stanford Institute for Theoretical Physics seminar organized by Xiaoliang Qi, Stanford, California (Feb 24, 2023).
- [Pitzer Center Theoretical Chemistry Seminar](#), Berkeley, California (Feb 22, 2023).
- [Redwood Seminar](#), Berkeley, California (Feb 22, 2023).
- [Würzburg Seminar on Quantum Field Theory and Gravity](#), Universität Würzburg, Online (Feb 7, 2023).
- [Fields Institute Seminar](#), University of Toronto, Ontario (Feb 3, 2023). [\[Recording\]](#)

Noncommuting charges: Bridging theory to experiment

- [Theoretical Physics Seminar Series](#), Australian Institute for Physics, Online (Aug 18, 2022). [\[Recording\]](#)
- [RQS Seminar](#), University of Maryland, Maryland (Aug 2, 2022).
- [Bristol QIT Seminars](#), University of Bristol, Online (Jun 9th, 2021).

An introduction to quantum thermodynamics

- [Mila Seminar](#), Mila, Online (Dec 1st, 2021).

INTERVIEWS & MEDIA RELATIONS

News

New Scientist feature on Phantom Codes. [Link](#).
Shayan Majidy awarded W.B. Pearson Medal. [Link](#).
Institute for Quantum Computing grad receives prestigious Faculty of Science medal. [Link](#).
Invited Panelist: Quantum + The Near Future. [Link](#).
Shayan Majidy wins prestigious Vanier Scholarship. [Link](#).
IQC student awarded Vanier Graduate Scholarship. [Link](#).
Graduate students recognized for excellence in research and scientific outreach. [Link](#).

Interviews

National Radio Interview: CBC's Quirks and Quarks. [Link](#).
Quantum + The Near Future Panel at [Quantum Connections 2024](#).

Media Interventions

IQC Achievement Award recipient Shayan Majidy shares research insights. [Link](#).
Quantum Q&A with Shayan Majidy. [Link](#).
Quantum Frontiers: Identical twins and quantum entanglement. [Link](#).
Quantum Frontiers: Mo' heights mo' challenges – Climbing mount grad school. [Link](#).
Quantum Frontiers: Building a Koi pond with Lie algebras. [Link](#).
Quantum Today: Bridging Quantum Thermodynamics Theory to Experiment. [Link](#).
Brief documentary on Unentangled by Ward1 Studios. [Link](#).
IQC Fireside Chat with Shayan Majidy - Sharing quantum science with a young audience. [Link](#).

TEACHING & MENTORING EXPERIENCE

Sessional Lecturer, University of Waterloo

Responsibilities included developing course content and syllabus, delivering lectures, developing and grading all assignments, managing teaching assistants, and holding office hours.

- *Introduction to Implementation of Quantum Information Processing* (Fall 2022)

Teaching Assistant, University of Waterloo

Responsibilities included leading discussion sessions, grading all assignments, and holding office hours

- *Quantum Information Processing Devices* (Winter 2020-22)
- *Electricity and Magnetism 1* (Winter 2021)
- *Introduction to Implementation of Quantum Information Processing* (Fall 2021)

Graduate Educational Developer, [Centre for Teaching Excellence](#) (Jan 2022—May 2022)

Responsibilities included teaching observations, facilitation of advanced workshops and microteaching sessions, and development of new workshops (10 hrs/week).

TA Workshop Facilitator, [Centre for Teaching Excellence](#) (Jan 2021—Dec 2021)

Responsibilities included facilitating graduate-student teaching workshops, preparing five to six workshops each term, and participating in monthly meetings to refine workshop offerings (30 hrs/term).

Teaching Accreditations

[Certificate in University Teaching](#)
[Fundamentals of University Teaching](#)

Mentoring

2 graduate students at Harvard, one at ETH, and one at the University of Waterloo.
One undergraduate student at Princeton, and 3 at the University of Waterloo

SELECTED FELLOWSHIPS & AWARDS

Fellowships

[Banting Fellowship \(\\$140,000\)](#), 2024–2026. Canada’s highest-valued Postdoctoral Fellowship. I was ranked 3rd out of 185 applicants in my field.

Prizes & Award

[W.B. Pearson Medal](#), 2025. Awarded to one PhD graduate annually in physics for a creative research in their doctoral thesis.

[Vanier Scholarship \(\\$150,000\)](#), 2021—2024. Canada’s highest-valued PhD scholarship.

[John Brodie Memorial Award \(\\$1,000\)](#), 2024. The Perimeter Institute’s highest valued PhD award, evaluated on research impact and independence.

[Institute for Quantum Computing’s Achievement Award \(\\$5,000\)](#), 2022. IQC’s highest valued PhD award, evaluated on “exceptional achievement in research.”

PROFESSIONAL SERVICE

Reviewer

Nature Journals (4): *Nature Communications* (1); *npj Quantum Information* (3).

Physical Review Journals (38): *Physical Review Letters* (3); *PRX Quantum* (8); *Physical Review Research* (2); *Physical Review A* (24); *Physical Review E* (1).

Conference review subreviewer: QIP 26 (2); QEC 25 (3); QCTIP 2025 (1); TQC 2024 (1).

Leadership

Founder, Unentangled, [Brief documentary](#)

Organizer, Raymond Laflamme’s 60th Birthday Conference, 2023

Seminar organizer, Quantum Steampunk Seminars, University of Maryland, 2021

Organizing committee, Canadian Graduate Quantum Conference 2020, University of Waterloo, 2020

Outreach

Panelist, [From TA to Course Instructor Workshop](#), 2023

Panel Facilitator, Developing Your Teaching Skills in Grad School, 2021

Session Chair, [University of Waterloo Teaching and Learning Conference](#), 2021

Facilitator, [USEQIP](#), 2019–2021

Facilitator, [Quantum Cryptography School for Young Students](#), 2018–2021

Facilitator, IQC Science Outreach, 2017–2021

Panelist, [Tech Under Twenty Expo](#), 2020

Facilitator, [EinsteinPlus](#), 2019

Committee Memberships

Member Physics GSA, 2019–2024

Member, Institute for Quantum Computing GSA, 2019–2024

Graduate student member, Faculty Committee on Student Appeals, 2020–2021

Co-President, Physics GSA,, 2019–2020

Executive member, Institute for Quantum Computing GSA, 2019–2020